Ranking Before Generation: Auditing Chunking, Retrieval Fusion, and Evidence Exposure in RAG

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ARTICLE INFO

Keywords:
retrieval-augmented generation
information retrieval
ranking
document chunking
reranking
robustness
evaluation

ABSTRACT

Retrieval-augmented generation is a multi-stage ranking system: documents are segmented, candidates are retrieved and fused, evidence is packed into a finite context, and a language model generates an answer. We study how chunk boundaries interact with these stages rather than treating chunking as an isolated preprocessing choice. First, we audit an archived Mistral-7B pilot on SQuAD 2.0 and HotpotQA. Replaying the exact token budgets shows that the generator truncates every SQuAD fixed-512 prompt and MiniLM truncates 93.9% of the corresponding corpus chunks during embedding. A corrected failure audit attributes 59.1% of recursive HotpotQA exact-match failures to incomplete retrieval rather than repairable generation. Second, a prospective study crosses three data seeds, dense, BM25, and hybrid retrieval, two embedding models, and four encoder-compatible chunkers over SQuAD, HotpotQA, and TechQA. A paired extension reranks BGE-hybrid candidates with an MS MARCO cross-encoder. Across chunkers, reranking raises HotpotQA AllHit@4 by 5.3–8.9 points but lowers TechQA AllHit@4 by 3.5–5.0 points, while increasing retrieval latency by 1.5×–6.0×. Controlled Qwen and Mistral generation likewise yields dataset-specific rankings, and none of 18 planned chunker contrasts survives Holm correction. Document hits, answer-bearing evidence, answer quality, and efficiency are distinct objectives. Reliable RAG evaluation must audit the complete retrieval–ranking–generation path and the evidence actually exposed to the generator.

1. Introduction

Retrieval-augmented generation (RAG) conditions a language model on external evidence rather than relying exclusively on parametric memory (Lewis, Perez, Piktus, Petroni, Karpukhin, Goyal, Küttler, Lewis, Yih, Rocktäschel, Riedel and Kiela, 2020). Most RAG systems index chunks instead of complete documents. A chunking policy therefore affects the retrieval unit, the amount of surrounding context, the number of indexed vectors, and the evidence that can fit into the generator prompt. These effects are coupled: a larger chunk may preserve discourse while exceeding an encoder or prompt budget, whereas a smaller chunk may fit easily but omit neighboring evidence.

We use a two-stage design. The first stage is a retrospective audit of an archived Mistral experiment in which the sampled corpus, embedding model, dense retriever, retrieval depth, prompt, generator, and evaluation code were fixed. The second is a prospectively specified extension that varies datasets, data seeds, retriever families, embedding models, and the generator while keeping each within-cell comparison paired. Together, the studies ask:

1. How do six chunker configurations compare on answer quality and supporting-document retrieval in the archived pipeline?
2. How uncertain are the observed paired differences on the sampled questions?
3. Do encoder and generator token limits change what each nominal chunking configuration actually exposes to the system?
4. Do observed chunker rankings persist across larger samples, a technical-support dataset, dense and lexical retrieval, two embedding models, cross-encoder reranking, and two controlled local generators?

This paper makes four contributions. First, it frames chunking as a control on the retrieval unit inside a multi-stage ranking pipeline and provides a controlled archival pilot with per-question predictions and separate answer and evidence-retrieval metrics. Second, it adds paired bootstrap intervals, randomization tests, and an overlap analysis showing that identical aggregate recall can conceal different missed questions. Third, it reconstructs the exact retrieved contexts and replays both tokenizer budgets. This audit changes the interpretation of the original experiment: some

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differences are inseparable from embedding and prompt truncation, and an earlier claim that most failures were “fixable” is not supported. Fourth, it executes a larger robustness extension across three datasets, three seeds, lexical, dense, fused, and cross-encoder reranked retrieval, two embedding models, and controlled local generation with Qwen2.5-1.5B and Mistral-7B. Reranking improves multi-hop support recovery on HotpotQA but harms both support-document and answer-evidence retrieval on TechQA. The extension therefore rejects both a dataset-independent chunker ranking and a domain-independent reranking benefit.

Code, configurations, per-question predictions, and audit outputs are included in the anonymous supplementary material.

2. Related Work

RAG and dense retrieval. RAG combines retrieval with conditional generation (Lewis et al., 2020); Dense Passage Retrieval established bi-encoder retrieval as a strong approach for open-domain QA (Karpukhin, Oguz, Min, Lewis, Wu, Edunov, Chen and Yih, 2020). Our implementation follows this broad design but does not use a DPR checkpoint: it embeds chunks with all-MiniLM-L6-v2, a Sentence-BERT-style encoder (Reimers and Gurevych, 2019) derived from MiniLM (Wang, Wei, Dong, Bao, Yang and Zhou, 2020), and searches normalized embeddings with inner product. The robustness extension also uses BGE-small-en-v1.5, an English embedding checkpoint from the BGE/FlagEmbedding family (Xiao, Liu, Zhang, Muennighoff, Lian and Nie, 2023), and compares dense retrieval with BM25 (Robertson and Zaragoza, 2009) and weighted reciprocal-rank fusion (Cormack, Clarke and Büttcher, 2009). A final stage applies a query–passage cross-encoder, following the standard neural reranking pattern (Nogueira and Cho, 2019).

Segmentation and retrieval granularity. Topical text segmentation predates neural retrieval; TextTiling, for example, detects subtopic boundaries from lexical cohesion (Hearst, 1997). Recent work treats retrieval granularity as an empirical variable. Dense X Retrieval shows that the choice of retrieval unit affects both retrieval and downstream QA (Chen, Wang, Chen, Yu, Ma, Zhao, Zhang and Yu, 2024). Semantic chunking attempts to align boundaries with embedding similarity, but its cost-benefit trade-off is inconsistent across evaluated settings (Qu, Tu and Bao, 2024). Late Chunking instead contextualizes token embeddings before pooling, illustrating that segmentation also interacts with when contextual representations are formed (Günther, Mohr, Williams, Wang and Xiao, 2024). Our study compares boundary rules within one conventional pre-chunking pipeline; it does not compare these broader embedding architectures.

RAG and QA evaluation. Knowledge-intensive evaluation should distinguish answer correctness from provenance and evidence retrieval (Petroni, Piktus, Fan, Lewis, Yazdani, De Cao, Thorne, Jernite, Karpukhin, Maillard, Plachouras, Rocktäschel and Riedel, 2021). Frameworks such as RAGAs and ARES separate context relevance, answer relevance, and faithfulness (Es, James, Espinosa Anke and Schockaert, 2024; Saad-Falcon, Khattab, Potts and Zaharia, 2024), while RAGTruth demonstrates that retrieval does not by itself prevent unsupported generation (Niu, Wu, Zhu, Xu, Shum, Zhong, Song and Zhang, 2024). Reading comprehension systems are also sensitive to distractors (Jia and Liang, 2017). These findings motivate our separate answer, supporting-document, and context-visibility measurements.

Domains and generators. SQuAD and HotpotQA cover Wikipedia reading comprehension and multi-hop reasoning, respectively (Rajpurkar, Jia and Liang, 2018; Yang, Qi, Zhang, Bengio, Cohen, Salakhutdinov and Manning, 2018). TechQA instead contains real technical-support questions grounded in product documentation (Castelli, Chakravarti, Dana, Ferritto, Florian, Franz, Garg, Khandelwal, McCarley, McCawley, Nasr, Pan, Pendus, Pitrelli, Pujar, Roukos, Sakrajda, Sil, Uceda-Sosa, Ward and Zhang, 2020). We use NVIDIA’s reduced RAG-oriented distribution of that dataset (NVIDIA Corporation, 2025) to test domain transfer. The prospective generation extension uses pinned local Qwen2.5-1.5B-Instruct (Yang, Yang, Zhang, Hui, Zheng, Yu, Li, Liu, Huang, Wei, Lin, Yang, Tu, Zhang, Yang, Yang, Zhou, Lin, Dang, Lu, Bao, Yang, Yu, Li, Xue, Zhang, Zhu, Men, Lin, Li, Tang, Xia, Ren, Ren, Fan, Su, Zhang, Wan, Liu, Cui, Zhang and Qiu, 2024) and Mistral-7B-Instruct-v0.3 (Jiang, Sablayrolles, Mensch, Bamford, Chaplot, de las Casas, Bressand, Lengyel, Lample, Saulnier, Lavaud, Lachaux, Stock, Le Scao, Lavril, Wang, Lacroix and El Sayed, 2023) checkpoints under the same retrieval, prompt, and decoding protocol. This controlled follow-up is distinct from the archived Mistral endpoint, whose runtime revision and full server flags were not retained.

Uncertainty and context use. Small NLP evaluations require paired tests and explicit uncertainty (Dror, Baumer, Shlomov and Reichart, 2018). Long input capacity is likewise not equivalent to robust context use: answer accuracy can depend on evidence position even within the nominal window (Liu, Lin, Hewitt, Paranjape, Bevilacqua, Petroni and Liang, 2024). Our audit addresses an earlier-stage issue—whether the retrieved evidence enters the model input at all—by replaying the exact truncation policy before interpreting retrieval-generation discrepancies.

3. Experimental Setup

3.1. Study Structure

We distinguish the *archival pilot* from the *prospective extension* throughout. The pilot is a single seed-42 Mistral run reconstructed from stored predictions; its audits cannot retroactively recover missing raw drafts or endpoint metadata. The extension was executed after the audit with pinned dataset and model revisions, retained raw generations and prompt-length traces, and configurations chosen to remove the pilot's most severe encoder-budget mismatch. Results from the extension test robustness rather than replacing or pooling with the archival measurements.

3.2. Datasets and Corpus Construction

SQuAD 2.0. We use only answerable validation questions (Rajpurkar et al., 2018). After a seed-42 shuffle, the loader takes a 500-example candidate pool and selects 60 questions round-robin across titles. Context paragraphs sharing a selected title are concatenated, yielding 35 article-level pseudo-documents. This title-balanced procedure is not a simple random sample of SQuAD 2.0, and none of SQuAD's unanswerable questions is evaluated.

HotpotQA. We select 30 seed-42 validation examples from the distractor configuration (Yang et al., 2018). Each question has two supporting titles and ten candidate documents. The retrieval corpus is the union of the question-level candidates: 300 documents. This construction differs materially from the 35-document SQuAD corpus.

TechQA. The extension adds NVIDIA TechQA-RAG-Eval, a reduced RAG-oriented distribution of IBM TechQA (Castelli et al., 2020; NVIDIA Corporation, 2025). It contains real technical-support questions, long product-support documents, and complete explanatory answers rather than mostly short Wikipedia spans. We filter out rows marked impossible or lacking contexts and sample evaluation questions after a seeded shuffle. Retrieval always searches the same closed corpus of 496 unique context documents from all answerable rows, rather than a question-conditioned set of gold documents. A filename identifies a gold support document.

Extension sample sizes. For retrieval-only evaluation, each of seeds 13, 21, and 34 selects 150 SQuAD questions from a 900-example candidate pool, 75 HotpotQA distractor questions, and 200 answerable TechQA questions. SQuAD retains the title-round-robin procedure; HotpotQA and TechQA use seeded shuffled samples. Both prospective generator runs use seed 42 and the same 60, 30, and 50 questions from SQuAD, HotpotQA, and TechQA, respectively. Thus the retrieval extension covers 425 questions per seed, while generation remains a smaller controlled two-checkpoint study.

3.3. Chunking Configurations

Chunk sizes are measured with the MiniLM tokenizer. We compare six configurations from four families:

- **Fixed windows:** targets of 128, 256, and 512 tokens with overlaps of 19, 38, and 76 tokens, respectively.
- **Recursive:** a 256-token target and 38-token overlap, using the ordered separators paragraph, newline, sentence-ending space, space, and character.
- **Sentence:** a blank English spaCy sentencizer greedily packs complete sentences to a 256-token target, without overlap.
- **Semantic:** adjacent sentences are split when MiniLM cosine similarity falls below 0.72 after a default 128-token minimum; a 256-token maximum forces a split. No overlap is used.

Because the algorithms and overlaps differ, the realized chunk-length distributions and chunk counts are part of each configuration. We therefore compare configurations, not a single isolated boundary variable.

Prospective extension. To fit MiniLM's 256-position encoder budget including two special tokens, the extension compares four configurations: fixed-128 (19-token overlap), fixed-254 (38-token overlap), recursive-254 (38-token overlap), and sentence-254 (no overlap). Sentence segments longer than the target are further split into token-bounded windows. Because WordPiece decode and re-encode are not always idempotent, every decoded window is rechecked and shortened until it contains at most 254 content tokens. An executable audit confirms zero target or encoder-limit violations in every extension cell. We omit fixed-512 and semantic-256: the former is known to exceed the archival encoder budget, whereas retaining semantic chunking would introduce an additional encoder-dependent boundary model into the MiniLM–BGE comparison. These choices narrow the extension to fixed, separator-aware, and sentence-aware boundaries under a common encoder budget.

3.4. Embedding and Retrieval

Chunks are encoded with all-MiniLM-L6-v2; embeddings are L2-normalized and indexed with FAISS inner-product search. The top four chunks are ranked for every question. MiniLM's SentenceTransformer configuration has a maximum sequence length of 256 encoded positions, including special tokens. The pipeline raises the separate tokenizer limit used to create chunks, but not the encoder limit, so overlong chunks are silently prefix-truncated during embedding. Section 4.3 quantifies this mismatch.

We report **DocCov@4**, the mean fraction of gold supporting documents represented among the four retrieved chunks, and **AllHit@4**, the fraction of questions for which all supporting documents are represented. These metrics coincide on SQuAD but not on HotpotQA. We also report **AnsVis@4**, whether the normalized reference-answer token sequence occurs inside at least one retrieved chunk. Normalized tokens must match on token boundaries, preventing short answers such as "IT" from matching inside words such as "internet." HotpotQA yes/no questions are excluded because literal label presence is not an evidence diagnostic. AnsVis is still conservative for paraphrased and long TechQA answers, but prevents a hit on an unrelated part of a long gold document from being mistaken for visible answer evidence. We omit the repository's original "Precision@4" from the main analysis because it mixes document identity with answer-string containment and counts multiple chunks from the same document separately.

Prospective retrieval extension. We cross the four extension chunkers with dense retrieval, BM25, and hybrid retrieval at $k = 4$. Dense retrieval uses either all-MiniLM-L6-v2 or BGE-small-en-v1.5; the BGE query is prefixed with its recommended English retrieval instruction (Xiao et al., 2023). The hybrid retrieves 20 candidates from each component and applies weighted reciprocal-rank fusion with weights 0.6 dense and 0.4 BM25. Since BM25 does not use the embedding model and both configurations use the same pinned MiniLM chunk tokenizer, its exactly matching result is shared rather than interpreted as two independent encoder runs. For each dataset–retriever–embedder–chunker cell, we report mean AllHit@4 and the sample standard deviation across the three data seeds.

Cross-encoder reranking extension. We fix BGE hybrid retrieval and rerank its top 20 fused candidates with `cross-encoder/ms-marco-MiniLM-L6-v2`, pinned to one model revision. The cross-encoder jointly scores each question–chunk pair; its four highest-scoring chunks replace the fused top four. Hybrid and reranked conditions use identical questions, corpora, chunkers, candidate-pool size, and data seeds, so each comparison is paired before the final top-k cutoff. We report AllHit@4, AnsVis@4, descriptive question-level gain/loss counts, and the ratio of mean end-to-end retrieval latency.

3.5. Prompt, Generator, and Stored Output

The generator is the Mistral 7B Instruct v0.3 checkpoint (Jiang et al., 2023), served through an OpenAI-compatible vLLM endpoint. The first completion uses temperature 0, at most 48 new tokens, and a complete-chat input limit of 1,024 Mistral tokens. The system message instructs the model to use only the context, copy the shortest supported answer span, omit explanations, and return exactly *unanswerable* when support is insufficient. HotpotQA passages include titles; SQuAD passages do not. Appendix A gives the exact messages.

If the formatted chat is too long, the implementation binary-searches for the longest prefix of the concatenated, rank-ordered context that keeps the complete chat at or below 1,024 tokens. It can remove later chunks or cut inside one. A first-line normalizer then strips answer labels and citations. Sentence-like or long drafts may trigger a second Mistral refinement call with at most 24 new tokens, followed by question-dependent compression rules. Consequently, the archived predictions are stored post-processed answers, not raw model generations. The artifact does not retain which examples invoked refinement or the intermediate draft; if refinement returns an empty string or *unanswerable*, the implementation retains the first draft.

The no-context comparator uses a different short-answer system prompt. It is therefore a useful descriptive comparator, not a formal lower bound on the contextual system.

Controlled local generator extension. The prospective generation study runs pinned Qwen2.5-1.5B-Instruct (Yang et al., 2024) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) checkpoints with BGE-small-en-v1.5, the same 0.6/0.4 hybrid retriever, the same retrieved chunk IDs, and the four extension chunkers. Mistral is loaded locally in float16 rather than through a paid or mutable hosted endpoint. Each model uses its native chat template and tokenizer, but receives the same instructions. Greedy decoding permits at most 96 new tokens within a 1,536-token complete-chat budget for SQuAD and HotpotQA; TechQA permits 512 new tokens because its references are substantially longer. SQuAD and HotpotQA use the archival extractive instruction; TechQA uses a concise-but-complete grounded-answer instruction because its references are often multi-sentence support resolutions. The implementation stores raw and normalized predictions, full and retained prompt lengths, generated-token counts, and truncation and length-stop indicators. It preserves multiline TechQA answers and does not invoke the archival endpoint's refinement call. Each model also has a no-context run with a different instruction; these are descriptive comparators, not formal lower bounds.

3.6. Answer Metrics and Statistical Analysis

We use SQuAD-style Exact Match (EM) and token F1: lowercase, remove punctuation and articles, normalize whitespace, and take the maximum over gold references. F1 is the primary answer endpoint. The repository's marginal intervals resample questions; this revision uses 20,000 draws for the reported percentile intervals. We additionally align systems by example ID, bootstrap paired F1 differences with 20,000 draws, and run two-sided paired sign-flip tests with 100,000 draws. The five post-hoc comparisons of `recursive_256` to the other chunkers are Holm-corrected within each dataset. All random procedures use seed 8677.

For each prospective generator, we apply the same 20,000-draw paired bootstrap and 100,000-draw sign-flip procedure to F1, comparing recursive-254 with the other three chunkers. The primary Holm family contains all 18 post-hoc tests across two generators, three datasets, and three comparators; within-dataset values are retained only for transparency. Marginal table intervals use 2,000 question-bootstrap draws. Retrieval-only means use three data seeds and are reported with sample standard deviations; with only three seeds, we avoid treating their dispersion as a precise population confidence interval.

For the reranking extension, deltas are computed within each seed before averaging. Gain/loss counts compare paired binary outcomes over seed–question instances. Because the seeded samples can overlap and there are only three seeds, we treat these counts and deltas as descriptive evidence rather than independent-trial significance tests.

The archival intervals express variation over sampled questions conditional on one recorded run; the prospective intervals are likewise conditional on one deterministic run per checkpoint. They do not capture generator-run or software variation. Question-level resampling also ignores dependence among SQuAD questions from the same title and may be optimistic. The three-seed retrieval summary directly measures data-sample variation but does not represent model or deployment variation.

3.7. Context-Budget and Failure Audits

We deterministically reconstruct the seed-42 corpora and all six chunk sets, map each archived retrieved chunk ID back to its text, and replay the Mistral chat template and prefix-truncation algorithm. Reconstructed chunk counts match the stored counts for every cell. For each question we record the full prompt length, retained context fraction, number of completely retained chunks, support-document coverage among those complete chunks, and whether a normalized SQuAD gold answer remains in the consumed context.

For predictions with EM$=0$, we use three coarse diagnostic categories. A SQuAD case is *evidence-limited* when no normalized gold answer survives in the reconstructed post-truncation context. A HotpotQA case is evidence-limited unless both supporting documents occur among completely retained chunks. Among the remainder, refusals and strict token-set containment mismatches are *response-form candidates*; other high-, partial-, and zero-overlap outputs are *answer-content mismatches*. The priority order is fixed and mutually exclusive. These automatic labels generate hypotheses; they are neither human causal judgments nor proof that a prompt or post-processor will repair a prediction.

System	EM	F1	F1 95% CI	DocCov@4	AllHit@4	Avg. tokens	# chunks
No context	13.3	29.2	[20.9, 38.1]	–	–	–	–
`fixed_128`	38.3	50.7	[39.9, 61.4]	**95.0**	**95.0**	125.3	631
`fixed_256`	38.3	56.7	[46.2, 66.9]	88.3	88.3	244.0	322
`fixed_512`	40.0	53.7	[43.1, 64.5]	90.0	90.0	472.2	164
`recursive_256`	**41.7**	**60.6**	[50.5, 70.5]	91.7	91.7	175.8	389
`sentence_256`	35.0	51.0	[40.2, 61.7]	86.7	86.7	223.9	302
`semantic_256`	**41.7**	56.9	[46.1, 67.3]	**95.0**	**95.0**	144.8	467

Table 1
Observed results on the fixed SQuAD 2.0 sample. EM, F1, DocCov@4, and AllHit@4 are percentage points. Intervals are question-level percentile-bootstrap intervals and do not measure run-to-run variation. Bold marks the largest observed chunker mean; it does not indicate statistical significance.

System	EM	F1	F1 95% CI	DocCov@4	AllHit@4	Avg. tokens	# chunks
No context	10.0	27.7	[16.2, 40.0]	–	–	–	–
`fixed_128`	20.0	36.5	[22.7, 51.2]	75.0	50.0	89.3	421
`fixed_256`	20.0	34.3	[19.8, 49.8]	75.0	50.0	114.2	313
`fixed_512`	20.0	35.7	[21.5, 50.8]	**76.7**	**53.3**	117.4	301
`recursive_256`	**26.7**	**42.6**	[26.8, 58.7]	**76.7**	**53.3**	113.6	313
`sentence_256`	23.3	37.0	[21.6, 53.0]	**76.7**	**53.3**	112.7	313
`semantic_256`	23.3	42.4	[27.6, 57.5]	**76.7**	**53.3**	97.2	363

Table 2
Observed results on the fixed HotpotQA sample. EM, F1, DocCov@4, and AllHit@4 are percentage points. Intervals are question-level percentile-bootstrap intervals and do not measure run-to-run variation. Bold marks the largest observed chunker mean; it does not indicate statistical significance.

Comparator	SQuAD 2.0		HotpotQA	
	ΔF1 [95% CI]	Holm p	ΔF1 [95% CI]	Holm p
`fixed_128`	+9.9 [+2.1, +18.4]	0.088	+6.1 [-2.3, +16.3]	0.809
`fixed_256`	+3.9 [-2.9, +11.1]	0.581	+8.3 [+0.2, +17.9]	0.627
`fixed_512`	+6.9 [+0.2, +14.4]	0.190	+6.8 [-2.0, +16.8]	0.809
`sentence_256`	+9.7 [+2.3, +17.8]	0.088	+5.6 [+0.0, +14.4]	0.998
`semantic_256`	+3.7 [-3.2, +11.0]	0.581	+0.2 [-6.5, +8.5]	1.000

Table 3
Paired comparisons defined as `recursive_256` minus each comparator. We resample aligned questions 20,000 times and compute two-sided paired sign-flip tests with 100,000 draws; p-values are Holm-adjusted across the five post-hoc contrasts within each dataset.

4. Results

4.1. Observed QA and Retrieval Results

On the fixed SQuAD sample, `recursive_256` has the highest observed F1 (60.6) and ties `semantic_256` on EM (41.7). On HotpotQA, recursive and semantic are nearly tied in F1 (42.6 versus 42.4). These are descriptive rankings: the F1 intervals are broad, especially for HotpotQA, where recursive spans 26.8–58.7.

DocCov@4 and AllHit@4 expose different aspects of multi-hop retrieval. Recursive HotpotQA DocCov@4 is 76.7, but both supporting documents are recovered for only 53.3% of questions. Four chunkers share both aggregate values while their observed F1 ranges from 35.7 to 42.6; aggregate document coverage alone does not resolve which questions or evidence spans differ.

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	Semantic-256	
Fixed-128	Hit	Miss
Hit	55	2
Miss	2	1

Table 4
Per-question SQuAD gold-article retrieval overlap. Both systems have 95.0% DocCov@4, but their three misses are not the same.

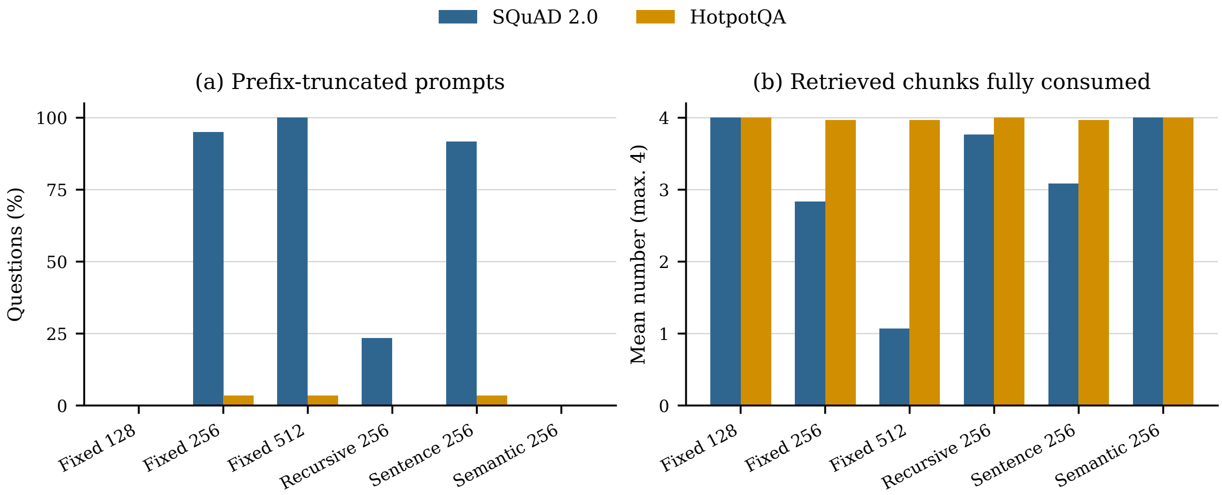

Figure 1: Replayed 1,024-token context budget. On SQuAD, fixed-256, fixed-512, and sentence-256 usually exceed the generator budget; HotpotQA documents are shorter. A chunk counted as fully consumed must survive the prefix truncation in its entirety.

4.2. Paired Uncertainty and Retrieval Identity

No recursive-versus-competitor F1 comparison has Holm-adjusted $p < .05$. The largest SQuAD observed differences are relative to fixed-128 and sentence-256, but both adjusted p-values are .088. Recursive exceeds semantic by only 3.7 F1 points on SQuAD (95% paired CI $[-3.2, 11.0]$) and 0.2 on HotpotQA ($[-6.5, 8.5]$). The experiment therefore does not statistically resolve a winning chunker.

Equal aggregate retrieval can also conceal distinct failures. Fixed-128 and semantic-256 each retrieve the SQuAD gold article for 57 of 60 questions. Table 4 shows that only one miss is shared: each system uniquely covers two questions missed by the other.

4.3. The Nominal Top Four Often Do Not Reach the Generator

The reviewer's concern about four 512-token chunks and a 1,024-token input is correct. All 60 SQuAD fixed-512 chats are truncated. Their full prompts average 2,362 Mistral tokens; after truncation, only 1.07 of four retrieved chunks are completely retained on average. Fifty-six questions retain chunk 1 plus part of chunk 2, and four retain the first two plus part of chunk 3; none retains all four. Fixed-256 and sentence-256 are also truncated for 95.0% and 91.7% of SQuAD questions. By contrast, fixed-128 and semantic-256 are never truncated. Recursive-256 is truncated for 14 of 60 questions, each retaining three complete chunks and part of the fourth.

The HotpotQA corpus contains much shorter documents: only one fixed-512 question is truncated, and recursive and semantic prompts are never truncated. The cross-dataset difference therefore reflects corpus construction as well as nominal chunk size.

The embedding stage introduces a second budget mismatch. MiniLM input length includes special tokens, leaving at most 254 content wordpieces. This truncates 284 of 322 SQuAD fixed-256 corpus chunks (88.2%) but only 7 of 389 recursive-256 chunks (1.8%), so even the nominally size-matched comparison changes encoder exposure. It also

Category	SQuAD	HotpotQA
Evidence limited	8 (22.9; 12.1–39.0)	13 (59.1; 38.7–76.7)
Form/refusal	19 (54.3; 38.2–69.5)	6 (27.3; 13.2–48.2)
Content mismatch	8 (22.9; 12.1–39.0)	3 (13.6; 4.7–33.3)

Table 5
Corrected audit of `recursive_256` errors (EM= 0). Cells show count (percentage; Wilson 95% CI). SQuAD evidence is checked after prompt truncation; HotpotQA requires both supporting documents. Form/refusal labels are hypotheses, not verified fixes.

truncates 154 of 164 SQuAD fixed-512 chunks (93.9%); only 12 of 301 HotpotQA fixed-512 chunks (4.0%) are affected. The SQuAD fixed-window results consequently combine boundary policies with encoder-side and generator-side truncation; they cannot identify a "large chunk" mechanism in isolation. Appendix Table 11 reports all configurations.

4.4. Corrected Failure Audit

The corrected audit reverses the original HotpotQA interpretation. Among 22 recursive predictions with EM= 0, 13 retrieve only one of two supporting documents and are evidence-limited. Six are response-form candidates after complete document retrieval, and three are residual answer-content mismatches. Thus the earlier claim that 16 of 22 (72.7%) were fixable format or refusal errors was produced by treating any nonzero support coverage as retrieval success. The stricter response-form share is 27.3% (Wilson 95% CI 13.2–48.2), and even this is not a repair rate.

For SQuAD, 8 of 35 recursive EM failures lack a normalized gold answer in the retained context, 19 are response-form candidates, and 8 are answer-content mismatches. Of four refusals that retrieve the gold article, only three retain a normalized answer string. This distinction matters because article-level retrieval does not guarantee that an answer-bearing span reaches the generator.

4.5. Prospective Multi-Setting Retrieval Extension

The retrieval extension does not reproduce one stable ranking. With MiniLM dense retrieval, the best observed AllHit@4 configuration is fixed-128 on SQuAD (97.1 ± 1.4), sentence-254 on HotpotQA (50.7 ± 4.8), and fixed-128 on TechQA (88.0 ± 3.0). With BGE dense retrieval, the corresponding choices are fixed-128 (97.8 ± 0.4), recursive-254 tied with sentence-254 (70.2 ± 8.9), and fixed-254 (88.8 ± 0.8). The 19.5-point increase in best dense HotpotQA AllHit@4 from MiniLM to BGE is larger than the within-encoder gaps among chunkers, illustrating that an embedding change can dominate the chunking comparison.

Hybrid retrieval changes the picture again. Fixed-128 is best observed on SQuAD for both MiniLM (98.9 ± 0.4) and BGE (99.1 ± 1.0), whereas HotpotQA favors fixed-254 with MiniLM (48.0 ± 5.8) and recursive-254 with BGE (58.7±3.5). On the fixed 496-document TechQA corpus, MiniLM hybrid ties fixed-128 and sentence-254 at 88.7 AllHit, while BGE hybrid favors fixed-128 (90.8 ± 1.3). BM25 is embedding-independent and is reported once conceptually even though the paired configuration files reproduce it.

Document identity and answer visibility select different chunkers. Sentence-254 has the highest observed AnsVis@4 in every dense and hybrid cell: for example, on TechQA it reaches 48.3 ± 5.3 with MiniLM dense and 49.2 ± 5.0 with BGE hybrid, far below the corresponding best document AllHit values. A gold-document hit therefore often retrieves the wrong region of a long support document. These are means and sample standard deviations over three seeds, not significance claims; the extension supports interaction, not a new universal winner. Table 7 reports every dense and hybrid setting in the appendix.

4.6. Cross-Encoder Reranking Trade-offs

Reranking is beneficial on the Wikipedia benchmarks but not on the technical-support domain. SQuAD document retrieval is already near ceiling, so AllHit@4 changes by at most 0.2 points; nevertheless, AnsVis@4 increases by 1.3–2.7 points across chunkers. For fixed-254, the reranker adds answer-visible evidence for 15 paired seed–question instances and removes it for 3, yielding the largest SQuAD net gain.

HotpotQA shows the clearest support-recovery benefit. AllHit@4 increases by 5.3–8.9 points across chunkers, and AnsVis@4 increases by 1.8–3.1 points. Under fixed-128, reranking converts 35 incomplete-support instances into complete hits while losing complete support on 15, a net gain of 20 among 225 paired seed–question instances. This is the setting in which a second-stage relevance model most clearly improves the multi-hop evidence set.

Dataset	Chunker	AllHit@4			AnsVis@4			Latency
		Hybrid	+Rerank	Δ	Hybrid	+Rerank	Δ	Ratio
SQuAD 2.0	`fixed_128`	99.1	99.1	+0.0	84.9	86.4	+1.6	2.6×
SQuAD 2.0	`fixed_254`	98.2	98.4	+0.2	84.4	87.1	+2.7	5.4×
SQuAD 2.0	`recursive_254`	98.7	98.7	+0.0	85.3	86.9	+1.6	6.0×
SQuAD 2.0	`sentence_254`	98.7	98.7	+0.0	93.3	94.7	+1.3	5.2×
HotpotQA	`fixed_128`	54.2	63.1	+8.9	62.2	65.3	+3.1	3.0×
HotpotQA	`fixed_254`	58.2	64.0	+5.8	64.4	67.6	+3.1	4.6×
HotpotQA	`recursive_254`	58.7	64.0	+5.3	64.4	67.6	+3.1	4.4×
HotpotQA	`sentence_254`	57.8	64.0	+6.2	68.0	69.8	+1.8	4.8×
TechQA	`fixed_128`	90.8	85.8	-5.0	12.5	12.0	-0.5	1.5×
TechQA	`fixed_254`	90.0	86.5	-3.5	21.7	19.3	-2.3	2.2×
TechQA	`recursive_254`	90.3	86.2	-4.2	20.7	19.7	-1.0	2.0×
TechQA	`sentence_254`	90.0	86.3	-3.7	49.2	46.7	-2.5	2.1×

Table 6
Paired BGE hybrid retrieval with and without cross-encoder reranking. Quality values and deltas are percentage points averaged over seeds 13, 21, and 34. Latency is the ratio of mean end-to-end retrieval time.

The direction reverses on TechQA. AllHit@4 decreases by 3.5–5.0 points and AnsVis@4 by 0.5–2.5 points across all four chunkers. Under fixed-128, only 6 instances gain a gold-document hit while 36 lose one. Under sentence-254, 16 instances gain visible answer evidence while 31 lose it. The MS MARCO-trained cross-encoder therefore does not provide a portable improvement for long technical-support documents. Reranking also raises mean retrieval latency by 1.5×–6.0×, so its quality gains are not free even where they are positive.

4.7. Controlled Local Generation Extension

Qwen yields dataset-dependent observed rankings. Its highest observed F1 is fixed-128 on SQuAD (66.58), sentence-254 on HotpotQA (53.37), and fixed-254 on TechQA (23.72). The no-context F1 values are 17.26, 21.91, and 13.97, respectively, so every observed RAG mean exceeds its descriptive no-context comparator while absolute TechQA overlap remains low. Because TechQA references are long resolutions, token F1 measures lexical coverage but does not establish factual completeness.

No Qwen recursive-versus-comparator contrast is significant: after the primary Holm correction, all nine Qwen adjusted p-values are 1.0. No SQuAD or HotpotQA prompt is truncated. Across the four TechQA RAG cells, fixed-254 truncates context for 2 of 50 questions and recursive-254 and sentence-254 for 1 each; fixed-128 truncates none. No Qwen output reaches its configured generation cap.

Mistral also yields dataset-dependent rankings. Fixed-128 has the highest observed SQuAD F1 (61.59), sentence-254 the highest HotpotQA F1 (49.64), and fixed-254 the highest TechQA F1 (26.50), compared with no-context F1 values of 27.78, 23.36, and 16.04. Its smallest raw paired value is for recursive-254 minus fixed-128 on SQuAD (-7.23 F1, 95% CI $[-13.62, -1.77]$, raw $p = .0156$), but the global Holm-adjusted value is .2817. None of the 18 planned contrasts across both generators survives the primary correction. Mistral truncates no SQuAD or HotpotQA prompt and five TechQA RAG prompts; no output reaches its configured generation cap. The complete two-model matrix, including marginal intervals, appears in Table 8.

5. Discussion

Across both studies, rankings are conditional on realized encoder and generator budgets, the retrieval stack, dataset, and generator. Document AllHit and answer visibility can select different chunkers, while equal aggregate coverage can hide different missed questions. Chunking recommendations should therefore identify the complete system and evidence endpoint, and preserve the text actually consumed after truncation.

The extension improves coverage with a new domain, larger multi-seed retrieval samples, two embedders, three retriever families, and two controlled local generators. It does not turn response-form diagnostics into proof of repairability: that claim requires a frozen prompt intervention evaluated on held-out questions. Appendix B states the prospective protocol and remaining interpretive cautions.

The reranking extension further shows why ranking integrity must be evaluated by domain and evidence endpoint. An MS MARCO cross-encoder improves complete multi-hop support on HotpotQA yet systematically drops gold documents and answer-bearing chunks on TechQA. One plausible explanation is domain and relevance-granularity mismatch: a generic passage ranker may favor locally query-similar excerpts over the long technical resolution containing the reference answer. Our experiment establishes the reversal, not its cause; testing that mechanism requires in-domain judgments or reranker adaptation. Reporting only an average benchmark gain would hide this deployment-relevant failure.

6. Conclusion

The evidence does not support a portable best chunk size. The pilot is confounded by encoder and prompt truncation, and its corrected HotpotQA audit finds incomplete retrieval more often than repairable generation. In the extension, observed winners change across datasets and system components, and no post-hoc contrast survives the 18-test correction. Cross-encoder reranking improves HotpotQA support recovery but harms TechQA retrieval and incurs substantial latency. Future RAG studies should match token budgets, retain consumed evidence and raw outputs, sample multiple data splits, and test both chunking and ranking claims across domains and model families.

7. Limitations

The archival component remains a pilot: it uses one seeded, answerable-only 60-question SQuAD sample, one 30-question HotpotQA sample, one embedding model, one dense retriever, and one 7B endpoint. One SQuAD question changes EM by 1.67 points and one HotpotQA question changes it by 3.33 points. Its question-bootstrap intervals do not measure run-to-run variation or SQuAD title clustering.

The encoder and generator token limits confound the chunker comparison, especially for SQuAD fixed-512. Supporting-document identity and normalized answer-token-sequence visibility are proxies for usable evidence, not proof that every reasoning step is present. The failure taxonomy is automatic and has no human validation or inter-annotator agreement. Raw first-pass generations, refinement decisions, and refined outputs were not stored, so the archived post-processed prediction cannot be decomposed retrospectively.

The extension is broader but not exhaustive. Three data seeds provide only a rough dispersion estimate, and each prospective generator is run once on the same seed and 60/30/50 questions. We test two embedders, three retriever families, and two generator checkpoints (1.5B Qwen and 7B Mistral), but family and size are confounded and this is not a systematic scale study. All datasets are English and only one technical-support domain is added. The reranking experiment tests one MS MARCO cross-encoder, one candidate-pool size, and one retrieval depth on a Colab T4; it does not isolate whether the TechQA reversal comes from domain shift, relevance granularity, or model capacity. We do not vary temperature, languages, or serving hardware. TechQA's complete-answer instruction differs from the extractive benchmark instruction, so absolute scores should not be compared across datasets. Token F1 under-rewards paraphrased long answers and does not measure factual completeness. Document identity is also only a proxy for passage-level evidence.

Finally, the archival configuration does not pin the exact vLLM version, Mistral model revision, dataset revisions, or complete server flags. The extension pins model and dataset revisions and stores prompt traces, but greedy decoding still does not make one run per checkpoint a multi-run generation evaluation. These constraints support conditional comparisons, not a general ranking of chunking methods.

8. Ethical Considerations

The study uses public QA benchmarks and does not collect new human-subject data. Its main risk is methodological: overgeneralized rankings could encourage practitioners to adopt a chunker without checking corpus, encoder, or context-budget interactions. We mitigate this risk by releasing per-question artifacts, reporting uncertainty, separating retrieved from consumed context, and explicitly withdrawing unsupported causal and "fixable failure" claims. The benchmarks and pretrained models may contain social or representational biases not assessed here. TechQA includes operational and security-related support material whose historical instructions may be obsolete. None of the evaluated systems should be treated as suitable for high-stakes or production technical support.

Declaration of Generative AI and AI-Assisted Technologies

During manuscript preparation, the authors used OpenAI Codex for language editing, code refactoring, experiment orchestration, and consistency checks. The authors reviewed and edited all generated material, verified reported results against retained artifacts, and take full responsibility for the content of the manuscript.

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A. Exact Prompts

The endpoint path uses chat messages, not the plain prompt used by the local `QAGenerator` class.

Contextual system message.

```
You are an extractive question answering assistant. Use only the provided context. Copy the shortest answer
span supported by the context. Do not explain your reasoning. If the answer is not fully supported, reply with
exactly 'unanswerable'.
```

Contextual user message.

```
Answer the following question using only the context.
Question: {question}
Context passages:
{ranked context}
Return only the answer text with no explanation.
```

No-context messages. The system says, "You are a concise question answering assistant. Return only a short answer phrase with no explanation." The user message is "Question: {question}" followed by "Return only the answer text." Because this instruction differs from the contextual one, the comparison changes both context availability and prompt wording.

Optional refinement system message.

```
You compress draft answers into the shortest exact answer span. Use the draft answer and the provided context.
Return only the minimal answer phrase. If the draft answer is unsupported, reply with exactly 'unanswerable'.
```

Optional refinement user message.

```
Question: {question}
Context passages:
{retained context, when it fits}
Draft answer: {draft answer}
Return only the final answer phrase.
```

The refinement call uses temperature 0 and at most 24 new tokens. If its complete chat exceeds 1,024 tokens, the context block is omitted. A simple answer of at most four word tokens and a pure number/quantity are not refined. Otherwise refinement is triggered by at least seven word tokens, brackets/parentheses/a colon, or copular and relation phrases such as "is," "owned by," or "from." An empty or *unanswerable* refinement is discarded and the draft is retained.

Deterministic post-processing. Normalization keeps the first line; removes leading citation numbers, answer labels, "the answer is," surrounding quotes, trailing citations, explanatory "implies" parentheticals, and terminal punctuation; and maps phrases containing "unanswerable," "not answerable," or "not supported" to *unanswerable*. The final compressor removes source parentheticals, then uses question-conditioned regular expressions to extract quantities for "how many"/population questions, dates for when/year questions, subjects for who questions, locations for where questions, and objects or complements of owned/founded/caused and common what/which constructions. If no rule matches, it returns the normalized string unchanged. The executable rules are in `src/chunkrag/generation.py`; raw drafts and refinement outputs were not archived.

Prospective local-generator prompts

For SQuAD and HotpotQA, Qwen and Mistral use the exact contextual and no-context messages printed above, without the optional refinement call. For TechQA, the contextual system message is:

```
You are a grounded question answering assistant. Use only the provided context. Give a concise but complete
answer containing the information needed to resolve the question. Do not explain your reasoning or add citations.
If the answer is not fully supported, reply with exactly 'unanswerable'.
```

The TechQA user message is:

```
Answer the following question using only the context.
Question: {question}
Context passages:
{ranked context}
Return only the final answer.
```

The TechQA no-context system message says, "You are a concise question answering assistant. Return a complete answer with no explanation." Its user message is "Question: {question}" followed by "Return only the final answer." If a contextual chat exceeds 1,536 tokens under the active model's native tokenizer and chat template, the implementation binary-searches for the longest prefix of the rank-ordered context that fits. Greedy decoding returns at most 96 new tokens for SQuAD and HotpotQA and 512 for TechQA. Normalization removes answer labels and citations from every output while preserving multiline TechQA content; question-conditioned compression is applied only to the extractive SQuAD and HotpotQA outputs.

B. Additional Discussion

The archived prompt already requests the shortest supported span, and the pipeline applies normalization, heuristic compression, and sometimes a refinement call. Containment mismatches or refusals therefore do not show that another prompt will help. A valid repair study should freeze an intervention developed away from the test questions, evaluate it on held-out examples, retain raw and final generations, and report paired uncertainty.

Likewise, document coverage is not usable-evidence proof. The overlap audit shows that equal aggregate recall can conceal different missed questions; HotpotQA requires both supporting documents; and TechQA often retrieves the correct long document without the answer-bearing region. Future evaluations should log post-truncation prompts, retained chunk spans, and support evidence rather than relying on document identity alone.

C. Supplementary Results and Audits

Configuration summaries. Table 9 separates settings preserved in the artifact from runtime details that were not recorded. The distinction prevents a configuration file from being treated as evidence for an unexecuted or incompletely specified run.

Prospective extension configuration. The exact generator revisions are `989aa7980e4cf806f80c7fef2b1adb7bc71aa306` for Qwen2.5-1.5B-Instruct and `c170c708c41dac9275d15a8fff4eca08d52bab71` for Mistral-7B-Instruct-v0.3. The JSON configurations similarly pin every dataset, chunk tokenizer, and embedding checkpoint revision.

Dataset	Embedder	Retriever	Best AllHit	Score	Best AnsVis	Score
SQuAD 2.0	MiniLM	dense	`fixed_128`	97.1 ± 1.4	`sentence_254`	84.7 ± 1.8
SQuAD 2.0	MiniLM	hybrid	`fixed_128`	98.9 ± 0.4	`sentence_254`	93.3 ± 3.1
SQuAD 2.0	BGE-small	dense	`fixed_128`	97.8 ± 0.4	`sentence_254`	89.1 ± 2.0
SQuAD 2.0	BGE-small	hybrid	`fixed_128`	99.1 ± 1.0	`sentence_254`	93.3 ± 1.3
HotpotQA	MiniLM	dense	`sentence_254`	50.7 ± 4.8	`sentence_254`	66.1 ± 5.2
HotpotQA	MiniLM	hybrid	`fixed_254`	48.0 ± 5.8	`sentence_254`	64.7 ± 5.3
HotpotQA	BGE-small	dense	`recursive_254`/`sentence_254`	70.2 ± 8.9/70.2 ± 8.9	`sentence_254`	78.6 ± 5.3
HotpotQA	BGE-small	hybrid	`recursive_254`	58.7 ± 3.5	`sentence_254`	72.3 ± 3.5
TechQA	MiniLM	dense	`fixed_128`	88.0 ± 3.0	`sentence_254`	48.3 ± 5.3
TechQA	MiniLM	hybrid	`fixed_128`/`sentence_254`	88.7 ± 3.0/88.7 ± 2.0	`sentence_254`	51.0 ± 4.6
TechQA	BGE-small	dense	`fixed_254`	88.8 ± 0.8	`sentence_254`	48.7 ± 4.5
TechQA	BGE-small	hybrid	`fixed_128`	90.8 ± 1.3	`sentence_254`	49.2 ± 5.0

Table 7
Best observed chunker under document AllHit@4 and normalized answer-token-sequence visibility (AnsVis@4) for each dense and hybrid setting. Scores are mean ± sample SD (percentage points) across seeds 13, 21, 34; ties are retained and list one score per tied chunker. HotpotQA yes/no questions are excluded from AnsVis because literal label presence is not an evidence diagnostic. AnsVis remains conservative for paraphrases and long answers. BM25 uses no embeddings and is reported once in the supplemental analysis.

Dataset	Model	No context	`fixed_128`	`fixed_254`	`recursive_254`	`sentence_254`
SQuAD 2.0	Qwen2.5-1.5B	17.3 [10.0, 24.9]	**66.6 [55.0, 77.1]**	61.5 [50.1, 71.9]	65.6 [54.5, 75.8]	65.3 [54.3, 75.6]
HotpotQA	Qwen2.5-1.5B	21.9 [8.9, 35.6]	50.3 [33.1, 68.6]	50.3 [33.5, 67.0]	47.0 [29.6, 64.1]	**53.4 [36.3, 70.0]**
TechQA	Qwen2.5-1.5B	14.0 [11.7, 16.4]	21.3 [17.0, 25.8]	**23.7 [19.7, 27.8]**	22.2 [17.9, 27.2]	20.4 [16.5, 24.8]
SQuAD 2.0	Mistral-7B	27.8 [19.2, 36.6]	**61.6 [49.9, 72.2]**	55.8 [45.1, 66.1]	54.4 [43.3, 65.3]	53.7 [42.3, 64.0]
HotpotQA	Mistral-7B	23.4 [13.6, 33.9]	45.4 [29.6, 61.4]	47.2 [31.9, 63.1]	45.7 [30.1, 61.7]	**49.6 [33.4, 65.7]**
TechQA	Mistral-7B	16.0 [13.7, 18.4]	24.2 [19.6, 29.1]	**26.5 [22.1, 31.1]**	23.8 [19.1, 28.7]	25.0 [20.7, 29.8]

Table 8
Generator F1 with no retrieved context and with BGE-small hybrid retrieval. Cells show F1 [marginal 2,000-draw question-bootstrap 95% CI] in percentage points; $n = 60/30/50$ for SQuAD/HotpotQA/TechQA. Bold marks the best observed RAG cell within each dataset and does not by itself imply statistical significance.

Component	Recorded setting
Data seed	42
SQuAD	validation; answerable only; 60 from a shuffled 500-example pool; 35 pseudo-documents
HotpotQA	distractor validation; 30 questions; 300-document union corpus
Embedding	`sentence-transformers/all-MiniLM-L6-v2`; normalized; max sequence length 256
Index	FAISS inner-product search
Retrieval	top-$k = 4$ dense chunks
Generator	`mistralai/Mistral-7B-Instruct-v0.3`; OpenAI-compatible vLLM endpoint
Decoding	temperature 0; 48 new tokens; no recorded seed; other endpoint defaults unrecorded
Input handling	complete chat at most 1,024 Mistral tokens; longest context prefix retained
Statistics	20,000 paired bootstrap draws; 100,000 sign-flip draws; seed 8677; Holm correction

Table 9
Settings recoverable from the committed configuration and code. Exact runtime package versions, model revisions, and full vLLM server flags were not stored with the prediction artifact.

Full context-budget audit. The special-token-inclusive MiniLM input exceeds its 256-position limit for 284 of 322 SQuAD fixed-256 chunks, 154 of 164 fixed-512 chunks, 7 of 389 recursive-256 chunks, and 23 of 302 sentence-256 chunks; fixed-128 and semantic-256 remain within the limit. The table's prompt audit uses the Mistral tokenizer and complete chat template, not the MiniLM tokenizer used to define nominal chunk sizes.

Per-question retrieval differences. Fixed-128 misses the SQuAD questions "What entity owns V/Line?", "Who was Ralph in charge of being at war with?", and "If two integers are multiplied and output a value, what is this expression set called?" Semantic-256 misses "What does critically tapered mean?", the Ralph question, and "What was the source of the mistake?" Thus the Ralph question is the only shared miss (miss-set Jaccard $= 1/5 = 0.20$).

Component	Executed setting
Retrieval data seeds	13, 21, and 34
Retrieval samples per seed	SQuAD 150 (900-candidate pool); HotpotQA 75; TechQA 200
Generation samples	seed 42 per checkpoint; identical SQuAD 60, HotpotQA 30, and TechQA 50 questions
Chunkers	fixed-128/19; fixed-254/38; recursive-254/38; sentence-254/0
Embeddings	all-MiniLM-L6-v2 and BGE-small-en-v1.5; normalized vectors
Retrievers	dense; BM25; hybrid weighted RRF (20 candidates; 0.6 dense/0.4 BM25); BGE hybrid followed by MS MARCO MiniLM cross-encoder reranking; top-$k = 4$
Reranking runtime	Google Colab Tesla T4; three seeds; paired 20-candidate reranking; retrieval latency includes the cross-encoder stage
Generators	Qwen2.5-1.5B-Instruct and Mistral-7B-Instruct-v0.3; BGE hybrid retrieval; identical retrieved chunk IDs
Decoding	greedy; 96 new tokens for SQuAD/HotpotQA and 512 for TechQA; complete chat at most 1,536 native-model tokens
Answer style	extractive for SQuAD/HotpotQA; concise complete answer for TechQA
Statistics	mean±sample SD over retrieval seeds; paired bootstrap/sign-flip tests for both generators; primary Holm family of 18
Stored output	per-question retrieval, raw/final generations, prompt lengths, truncation flags
Revisions	dataset, embedding, Qwen, and Mistral revisions pinned in the committed JSON configs

Table 10
Executed prospective extension. Unlike the archival pilot, model and dataset revisions and per-question generator traces are preserved.

	SQuAD 2.0			HotpotQA	
System	Embed cut (%)	Prompts cut (%)	Full chunks	Prompts cut (%)	Full chunks
`fixed_128`	0.0	0.0	4.00	0.0	4.00
`fixed_256`	88.2	95.0	2.83	3.3	3.97
`fixed_512`	93.9	100.0	1.07	3.3	3.97
`recursive_256`	1.8	23.3	3.77	0.0	4.00
`sentence_256`	7.6	91.7	3.08	3.3	3.97
`semantic_256`	0.0	0.0	4.00	0.0	4.00

Table 11
Replayed token-budget audit. “Embed cut” is the percentage of corpus chunks whose special-token-inclusive MiniLM input exceeds its maximum sequence length of 256. “Prompts cut” reports questions whose full chat exceeds 1,024 Mistral tokens; “full chunks” is the mean number of top-four chunks completely retained after prefix truncation.

Fine diagnostic categories. For recursive-256, the corrected SQuAD counts among 35 EM failures are: answer string not visible, 8; refusal after visible evidence, 3; gold token set contained with extras, 14; prediction token set contained in a gold answer, 2; high lexical overlap, 3; partial lexical overlap, 3; and zero lexical overlap, 2. The HotpotQA counts among 22 failures are: incomplete supporting-document retrieval, 13; refusal after both documents are retrieved, 4; gold token set contained with extras, 2; partial lexical overlap, 1; and zero lexical overlap, 2. Category definitions and all per-question inputs are included in the audit JSON.